ModelIt!

Teacher's Guide

Can We Reverse Aging?

Modeling Telomere Dynamics, Cellular Senescence, and Epigenetic Reprogramming

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS1-4

Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-component model to simulate the interacting mechanisms of cellular aging, testing how interventions targeting different aging pathways affect the trajectory of biological decline and healthspan.
Disciplinary Core Idea	**LS1.B Growth and Development of Organisms** — Students model how the cellular mechanisms that maintain organisms — cell division, tissue repair, stem cell function — progressively fail with age due to telomere shortening, senescence accumulation, and epigenetic drift.
Crosscutting Concept	**Stability and Change** — Students analyze how the stability of biological systems is maintained through youth by repair mechanisms and disrupted in aging by the accumulation of damage that overwhelms repair capacity, identifying the tipping points where decline accelerates.

CAST Testing Alignment

MULTIPLE CHOICE:

A 65-year-old patient receives senolytic therapy that eliminates 50% of their senescent cells. Based on the aging model, which downstream effect would be observed FIRST and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why the rate of aging accelerates after approximately age 60, referencing the senescence-inflammation feedback loop and at least three other model components. Why is this acceleration not simply a linear continuation of earlier aging?

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Aging was once considered an inevitable, immutable law of biology — as natural and inescapable as gravity. But over the past two decades, research has revealed that aging is driven by specific, identifiable molecular mechanisms that can be manipulated, slowed, and in some cases reversed. The question has shifted from 'why do we age?' to 'which aging mechanisms can we target?' and ultimately 'how much of aging is reversible?'

Telomere Biology and the Hayflick Limit

In 1961, Leonard Hayflick discovered that human cells can only divide approximately 40-60 times before permanently stopping — the Hayflick limit. We now know this limit is determined by telomere length. Telomeres — protective DNA caps on chromosome ends — shorten by 50-200 base pairs per division because DNA polymerase cannot fully replicate the very end of a linear chromosome (the end-replication problem). When telomeres reach a critical length (approximately 4-5 kilobases), cells activate DNA damage checkpoint responses that trigger senescence or apoptosis. The enzyme telomerase can rebuild telomeres and is active in stem cells, germ cells, and most cancer cells, but is repressed in most somatic cells, making telomere shortening a progressive, irreversible clock in normal tissues.

Cellular Senescence and SASP

Senescent cells were once thought to be harmless — simply cells that stopped dividing. Research by Judith Campisi and others revealed that senescent cells are actively harmful: they secrete a complex mix of inflammatory cytokines, growth factors, and matrix-degrading enzymes called the senescence-associated secretory phenotype (SASP). SASP signals damage neighboring cells, recruit inflammatory immune cells, degrade tissue structure, and can paradoxically promote cancer in nearby cells. The accumulation of senescent cells creates a toxic microenvironment that drives chronic inflammation, tissue deterioration, and organ dysfunction. In mice, selectively eliminating senescent cells with senolytic drugs reversed age-related decline, improving physical function by 25-35% and extending healthy lifespan.

Epigenetic Aging and Reprogramming

The epigenetic clock, developed by Steve Horvath in 2013, uses DNA methylation patterns at specific genomic sites to predict biological age with remarkable accuracy (within 3-4 years). This clock reveals that biological age can differ from chronological age and is influenced by lifestyle, disease, and interventions. More dramatically, Shinya Yamanaka's discovery that four transcription factors (Oct4, Sox2, Klf4, c-Myc) can reprogram adult cells back to a pluripotent stem cell state suggests that the epigenetic program of aging is reversible. Partial reprogramming — activating Yamanaka factors briefly without fully reverting cells to stem cells — has rejuvenated aged cells in mice, restoring youthful gene expression, epigenetic patterns, and tissue function. Altos Labs, funded with $3 billion, is pursuing this approach for human therapies.

Caloric Restriction and Longevity Pathways

Caloric restriction — reducing food intake by 20-40% while maintaining nutrition — is the most reproducible life-extending intervention across species, from yeast to primates. CR activates cellular stress response pathways (AMPK, sirtuins, autophagy) while suppressing growth-promoting pathways (mTOR, insulin/IGF-1). This shifts cells from a growth mode to a maintenance mode: increasing DNA repair, clearing damaged proteins and organelles, reducing oxidative stress, and slowing telomere erosion. The CALERIE trial in humans showed that 2 years of 12% caloric restriction slowed biological aging by 2-3% as measured by the DunedinPACE epigenetic clock. Drugs that mimic caloric restriction effects without reducing food intake — like rapamycin, metformin, and NAD+ precursors — are now in clinical trials.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the aging system: chronological age, oxidative stress, caloric intake, telomere length, senescent cell burden, chronic inflammation, stem cell function, epigenetic age, mitochondrial function, and healthspan index.

E **Establish Relationships**

Students map the aging network showing how telomere shortening triggers senescence, senescent cells drive inflammation, inflammation damages stem cells, and stem cell failure accelerates tissue decline, with oxidative stress and caloric intake as key modulators.

V **Visualize & Model**

Students build a ten-component computational model showing the interconnected aging mechanisms with critical feedback loops between senescence, inflammation, and stem cell exhaustion.

E **Evaluate Outcomes**

Students run normal aging, caloric restriction, and senolytic therapy scenarios to identify which aging mechanisms are most amenable to intervention and when interventions have the greatest impact.

R **Revise & Extend**

Students add autophagy rate, NAD+ levels, or immune senescence to model additional aging pathways and evaluate multi-target anti-aging strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Right now, inside every one of your cells, a molecular clock is ticking. It shortens with every cell division. When it runs out, the cell stops working and starts poisoning its neighbors. This process is called aging, and for the first time in history, we are learning how to slow it down — and maybe reverse it."*

Watch For

Show cover image. Ask: how old do you think you will live to be? How many of those years do you want to be healthy? Introduce the concept of healthspan versus lifespan.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the biology of aging — not as an inevitable decline but as a network of specific mechanisms that science is learning to manipulate. You will discover what drives aging, what accelerates it, and what might slow or reverse it."*

Watch For

Have students read objectives. Pre-teach 'telomere' with a shoelace analogy: telomeres are like the plastic tips (aglets) on shoelaces — they protect the ends, and when they wear away, the chromosome frays. Pre-teach 'senescence' as zombie cells — alive but not functioning, and actively harming neighbors.

Slide 3: Big Question

3 min

Say: *"The average American lives to 78 but only has about 63 healthy years. That is 15 years of decline, disease, and disability. Is that gap inevitable, or can biology offer a better deal?"*

Watch For

Quick-write: Students write what they think causes aging at the cellular level and whether they believe it can be reversed. Save to compare after the model exploration.

Slide 4: LEVER Framework

2 min

Say: *"We will Label ten aging mechanisms, Establish the feedback loops that drive accelerating decline, Visualize the aging network, Evaluate interventions that target different mechanisms, and Refine by adding autophagy and immune aging."*

Watch For

Preview each LEVER step. Emphasize that aging research has transformed from descriptive (what happens) to mechanistic (why it happens) to interventional (what we can do about it).

Slide 5: Activity 1: Components

8 min

Say: *"Three inputs drive aging from outside the cell: time itself, oxidative damage, and how much you eat. Seven processes inside the cell determine how fast you age biologically. Sort them and identify which components you can control through lifestyle choices."*

Watch For

Students sort external versus internal components. Discuss: caloric intake is external and controllable — what does this mean for your power over your own aging rate? Identify which internal components are affected by lifestyle.

Slide 6: Activity 2: Connections

8 min

Say: *"The key to understanding aging is the feedback loops. Senescent cells cause inflammation. Inflammation damages more cells. Damaged cells become senescent. This cycle accelerates after age 60. Find this loop in your model and trace how it affects everything downstream."*

Watch For

Students draw arrows with special attention to the senescence-inflammation positive feedback loop. Discuss: what makes this loop so dangerous? Why does aging seem to accelerate rather than progress at a constant rate?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three scenarios: normal aging from 20 to 80, the same person with 30% caloric restriction starting at 40, and senolytic drug treatment at age 65. Record which aging markers change most and when interventions have the biggest impact."*

Watch For

Students run all three scenarios and compare aging trajectories. Key discussion: the senolytic scenario shows healthspan improvement at 65 — does this mean aging is reversible? Or are we just removing a source of active damage?

Slide 8: Discoveries

5 min

Say: *"Your model reveals that aging is not a single process but a network, that the senescence-inflammation feedback loop is the critical accelerator, that caloric restriction slows the entire network, and that senolytics show aging is at*

least partially maintained rather than purely accumulated damage."

Watch For

Class discussion of discoveries. Ask: what is the difference between damage that accumulates irreversibly and damage that is actively maintained by senescent cells? Why does this distinction matter for anti-aging therapy?

Slide 9: STEM Challenge

12 min

Say: *"A biotech company has a combination anti-aging therapy. Design the clinical trial: who do you test it on, what do you measure, how long does the trial need to run, and what ethical issues must you address when the goal is not treating disease but slowing aging itself?"*

Watch For

Groups design clinical trial protocols with target populations, biomarker panels, endpoints, and ethical frameworks. Present and evaluate: which trial design would generate the most convincing evidence? Which best addresses equity concerns?

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Chronological Age, Oxidative Stress Level, Caloric Intake Level
- **INTERNAL:** Telomere Length, Senescent Cell Burden, Chronic Inflammation Level, Stem Cell Function, Epigenetic Age, Mitochondrial Function, Healthspan Index

Relationships

- **Chronological Age to Telomere Length:** NEGATIVE (-) — Each cell division shortens telomeres by 50-200 base pairs, and the cumulative divisions over a lifetime progressively erode telomeric DNA. More years mean more divisions and shorter telomeres.
- **Oxidative Stress Level to Telomere Length:** NEGATIVE (-) — Reactive oxygen species directly damage telomeric DNA, which is particularly vulnerable to oxidative damage due to its guanine-rich sequence. Oxidative stress accelerates telomere shortening beyond the rate caused by cell division alone.
- **Telomere Length to Senescent Cell Burden:** NEGATIVE (-) — When telomeres reach critically short length, cells activate DNA damage checkpoint responses that trigger permanent growth arrest (senescence). Shorter average telomere length means more cells have crossed this threshold.
- **Senescent Cell Burden to Chronic Inflammation Level:** POSITIVE (+) — Senescent cells secrete SASP — inflammatory cytokines, growth factors, and proteases — that create chronic, low-grade systemic inflammation. More senescent cells produce more inflammatory signals.
- **Chronic Inflammation Level to Stem Cell Function:** NEGATIVE (-) — Chronic inflammation damages stem cell niches, activates stem cells prematurely (exhausting the reserve), and impairs differentiation. Inflammatory signaling transforms the stem cell microenvironment from supportive to hostile.
- **Stem Cell Function to Healthspan Index:** POSITIVE (+) — Functional stem cells are essential for tissue repair, immune system renewal, and organ maintenance. As stem cell function declines, tissues accumulate damage they cannot repair, directly reducing physical function and disease resistance.
- **Caloric Intake Level to Oxidative Stress Level:** POSITIVE (+) — Higher caloric intake increases metabolic rate and mitochondrial electron transport chain activity, producing more reactive oxygen species as metabolic byproducts. Caloric restriction reduces this oxidative burden at the source.
- **Mitochondrial Function to Healthspan Index:** POSITIVE (+) — Efficient mitochondria produce adequate ATP energy for cellular function while minimizing ROS waste production. As mitochondrial function declines with age, cells produce less energy and more oxidative damage, creating a self-reinforcing decline cycle.

Key Discoveries

- Aging is not a single process but a network of interacting mechanisms — telomere shortening triggers senescence, senescent cells drive inflammation, inflammation damages stem cells, stem cell failure impairs tissue repair, and impaired repair accelerates further senescence, creating compounding decline
- The senescence-inflammation feedback loop is the critical accelerator of aging — once Senescent Cell Burden exceeds a threshold (approximately 15-20% of cells), their collective inflammatory secretions push neighboring cells into senescence faster than the body can clear them, creating exponential deterioration
- Caloric restriction slows aging by reducing oxidative stress at the source — less metabolic activity means less mitochondrial ROS production, slower telomere erosion, less DNA damage, and fewer cells entering senescence, effectively slowing the entire aging network simultaneously

- Senolytics demonstrate that aging is at least partially reversible — removing accumulated senescent cells reduces chronic inflammation, reactivates stem cell function, and improves tissue repair, showing that some age-related decline is maintained by the presence of damaged cells rather than caused by irreversible damage

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Aging is simply wear and tear that cannot be reversed"	While some aging damage is irreversible (accumulated mitochondrial DNA mutations, certain protein crosslinks), research has shown that major components of aging are actively maintained rather than passively accumulated. Senescent cells continuously secrete inflammatory signals that drive ongoing tissue damage — remove the senescent cells and the damage stops and partially reverses. Epigenetic reprogramming can restore youthful gene expression patterns to aged cells. These findings prove that aging is not simply accumulated wear but an active biological process that can be intervened upon.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We are close to achieving immortality"	Current anti-aging research aims to extend healthspan — the years lived in good health — not to achieve immortality. Even the most optimistic researchers estimate that healthspan could be extended by 10-30 years with combination therapies. The fundamental constraints of biology — entropy, the second law of thermodynamics, the accumulated complexity of damage across trillions of cells — make literal immortality physically implausible. The realistic and valuable goal is compressing morbidity: reducing the period of age-related decline and disease at the end of life.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Aging affects everyone equally"	The rate of biological aging varies enormously between individuals due to genetics, environment, lifestyle, and socioeconomic factors. People in poverty age biologically faster due to chronic stress, poor nutrition, environmental toxin exposure, and limited healthcare access. Studies show that the biological age gap between affluent and disadvantaged communities can exceed 10 years by middle age. Any discussion of anti-aging interventions must address this existing inequity — longevity therapies that are only accessible to the wealthy would widen an already unconscionable gap.